WORKSHEET

Chapter 6: Finding indefinite integrals 1

1 Find the indefinite integral of each of the following giving answers with positive indices.

a x^2

b $3x^5$

c $4x^3$

d $-2x^3$

e $x^{\frac{1}{3}}$

f $\frac{2x^4}{x^3}$

g $\frac{3}{x^2}$

h $\sqrt{x}$

i $\sqrt[3]{x}$

j $\sqrt[5]{x}$

2 Find the indefinite integral of each of the following.

a $\sin(x)$

b $\sin(3x)$

c $-\sin(7x)$

d $\cos(x+1)$

e $\sin(2x-3)$

f $\cos(2x-1)$

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g $\sin(\pi - x)$

h $\cos(x + \pi)$

i $4\sin\left(\frac{x}{2}\right)$

j $-\sin(3 - x)$

3 Find the antiderivative of each of the following.

a e^{2x}

b e^{4x}

c e^{-x}

d e^{5x}

e e^{-2x}

f e^{4x+1}

g $-3e^{5x}$

h $4e^{-3x}$

i $e^{7x} - 2$

j $e^{x-3} + x$

4 Find the antiderivative of each of the following.

a $(x + 1)^4$

b $(5x - 1)^9$

c $(4 + 3x)^4$

d $(7x + 8)^{12}$

e $(1 - x)^6$

f $\sqrt{2x - 5}$

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g $\dfrac{1}{2(4x-5)^3}$

h $\sqrt[3]{4x+3}$

i $(2-x)^{-\frac{1}{2}}$

j $\sqrt{(5x+2)^5}$

Answers

1 **a** $\frac{x^3}{3} + c$

b $\frac{x^6}{2} + c$

c $x^4 + c$

d $-\frac{x^4}{2} + c$

e $\frac{3x^{\frac{4}{3}}}{4} + c$

f $x^2 + c$

g $-\frac{3}{x} + c$

h $\frac{2\sqrt{x^3}}{3} + c$

i $\frac{3\sqrt[3]{x^4}}{4} + c$

j $\frac{5\sqrt[5]{x^6}}{6} + c$

2 **a** $-\cos x + c$

b $-\frac{1}{3}\cos 3x + c$

c $\frac{1}{7}\cos 7x + c$

d $\sin(x + 1) + c$

e $-\frac{1}{2}\cos(2x - 3) + c$

f $\frac{1}{2}\sin(2x - 1) + c$

g $\cos(\pi - x) + c$

h $\sin(x + \pi) + c$

i $-8\cos\left(\frac{x}{2}\right) + c$

j $-\cos(3 - x) + c$

3 **a** $\frac{e^{2x}}{2} + c$

b $\frac{e^{4x}}{4} + c$

c $-e^{-x} + c$

d $\frac{e^{5x}}{5} + c$

e $\frac{-e^{-2x}}{2} + c$

f $\frac{e^{4x+1}}{4} + c$

g $\frac{-3e^{5x}}{5} + c$

h $\frac{-4e^{-3x}}{3} + c$

i $\frac{e^{7x}}{7} - 2x + c$

j $e^{x-3} + \frac{x^2}{2} + c$

4 **a** $\frac{(x + 1)^5}{5} + c$

b $\frac{(5x - 1)^{10}}{50} + c$

c $\frac{(4 + 3x)^5}{15} + c$

d $\frac{(7x + 8)^{13}}{91} + c$

e $-\frac{(1 - x)^7}{7} + c$

f $\frac{\sqrt{(2x - 5)^2}}{3} + c$

g $-\frac{1}{16(4x - 5)^2} + c$

h $\frac{3\sqrt[3]{(4x + 3)^4}}{16} + c$

i $-2(2 - x)^{\frac{1}{2}} + c$

j $\frac{2\sqrt{(5x + 2)^7}}{35} + c$

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